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From Interpolation to Physics: A Reaction–Diffusion Framework for Grid-Aware Stochastic Field Generation

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Abstract

Stochastic field generation plays a central role in modeling subsurface properties such as permeability, porosity, and lithofacies. The traditional approach in the petroleum industry—Kriging and its extension, Sequential Gaussian Simulation (SGS)—treats the field as a statistical interpolation problem based on limited data. As such, it often struggles to capture sharp transitions, long-range connectivity, and directional features—key geological heterogeneities that influence flow and arise from natural depositional and diagenetic processes.

We propose a forward-modeling algorithm that mimics the formation of geological parameter fields by evolving a stochastic system over time. The method solves a reaction–diffusion equation with additive Gaussian noise. The reaction term incorporates constraints from geostatistical data and geologic structure, while the diffusion term promotes spatial coherence. Gaussian noise introduces heterogeneity and simulates the inherent randomness of natural systems. By calibrating noise strength and diffusion coefficients, we generate spatially correlated fields whose variogram and value distribution (histogram) emerge naturally and can be monitored dynamically. In contrast to traditional methods based on sequential simulation and Kriging-based interpolation, our approach embeds these geostatistical targets directly into the dynamics of a stochastic partial differential equation. Once the desired spatial properties are achieved, the simulation is stopped.

One of the key advantages is that stochastic parameter fields are generated directly on simulation grids—such as corner-point grids or unstructured meshes—rather than mapped after grid creation. Since both reservoir simulation and reaction–diffusion are governed by evolution equations, the transition between parameter generation and flow simulation becomes seamless. This also enables local control of heterogeneity near faults, boundaries, and well zones by adjusting diffusion coefficients. The approach supports a new modeling paradigm: generate the simulation grid first, then evolve the parameter field within it, reversing the conventional workflow.

We demonstrate the method through three numerical experiments. First, we demonstrate that the empirical variogram of the simulated field converges toward the prescribed target model as the field evolves. Second, we incorporate hard conditioning data using reaction terms that pull the solution toward specified values at target locations. Third, we embed sharp geologic features—such as streaks or barriers

—by applying spatial masks that locally modify the field. While the first two cases use regular grids, the third applies the method on corner-point grids to ensure compatibility with standard reservoir modeling workflows. The results show that this reaction–diffusion framework offers a flexible, interpretable, and physics-consistent alternative to SGS, producing geologically realistic parameter fields directly on simulation grids.

Introduction

Modeling geological heterogeneity is central to reservoir simulation, as it governs fluid flow, sweep efficiency, and uncertainty in decision-making. Properties such as permeability, porosity, and facies arise from complex depositional and diagenetic processes, exhibiting spatial patterns that are challenging to capture. Traditional geostatistical workflows are built upon kriging-based interpolation, with methods like Sequential Gaussian Simulation (SGS) (Deutsch and Journel, 1998; Kelkar and Perez, 2002; Deutsch, 2002; Pyrcz and Deutsch, 2014) extending kriging by introducing stochastic variability. These approaches, implemented in tools like GSLIB, are computationally efficient and widely applied in reservoir and geologic modelings.

Up to now, kriging remains one of the most established geostatistical interpolation techniques for continuous spatial fields, offering the best linear unbiased estimate under a specified variogram model (Stein, 1999; Pyrcz and Deutsch, 2014). Its extensions, including SGS, provide efficient ways to condition to well data and incorporate stochastic variability into realizations. These methods are computationally attractive and widely adopted in practice, though their reliance on stationary, local interpolation frameworks makes them less effective in capturing long-range connectivity or abrupt transitions in geological facies (Guardiano and Srivastava, 1993).

To overcome some of these limitations, object-based stochastic modeling was developed to directly represent channels, bars, or lobes as spatial objects. Early applications demonstrated that realistic geometries and connectivity in fluvial and deltaic systems could be preserved by explicitly embedding geobody shapes, orientations, and dimensions (Bratvold et al., 1994; Deutsch and Wang, 1996; Holden et al., 1998). Industry tool STORM extended this approach to 3D reservoir modeling, enabling geologists and engineers to specify depositional analogs and incorporate geological expertise into reservoir realizations in a natural way.

Multiple-Point Geostatistics (MPS) further advanced the field by replacing two-point variogram statistics with higher-order spatial templates extracted from training images (Strebel, 2002; Mariethoz and Caers, 2015). This allowed the reproduction of channels, lobes, or networks that cannot be captured by variograms alone. Extensions of MPS have also been applied in history matching under training image–based geological constraints (Caers, 2003). By leveraging training images, MPS provides a powerful mechanism for introducing geologically plausible structures into simulations, complementing the more geometric nature of object-based models.

An alternative, biologically inspired perspective on spatial pattern formation comes from reaction–diffusion (RD) systems, originally introduced by Turing (1952) to explain the spontaneous emergence of natural patterns such as animal coat markings or shell spirals. RD systems model the interplay between local activation (reaction) and long-range inhibition (diffusion), producing self-organized structures without the need for explicit templates. Since Turing's original work, RD models have been widely applied across physics, chemistry, ecology, and developmental biology (Kondo and Miura, 2010; Miura and Maini, 2004; Doelman et al., 2018). Beyond biology, RD has found uses in materials science and engineering—for example, in designing materials with tunable properties, 3D printing, water filtration membranes, and atomic-scale device fabrication. These successes across multiple disciplines demonstrate RD's power as a general framework for pattern formation.

Despite the promise of RD systems for pattern generation, they have rarely been applied in reservoir modeling. Our recent work (Wu, 2025 c) introduced RD dynamics as a foundation for automatic history

matching, where the time evolution of subsurface parameters was governed by a denoising PDE with adjoint-derived gradients and nonlinear diffusion terms. That work demonstrated that RD systems could preserve geological structure while guiding the model toward better alignment with observed production data.

In the present study, motivated by this versatility, we propose extending RD concepts into reservoir modeling, where geological heterogeneity also arises from underlying processes and can be simulated through dynamic, physics-inspired evolution rather than interpolation. We extend the RD concept to forward modeling by developing a stochastic field generator that evolves over time via a reaction–diffusion equation. The gradient of a variogram-based mismatch is introduced into the RD system as a feedback source term, guiding the spatial structure of the field. Meanwhile, diffusion coefficients are adaptively adjusted to control spatial coherence, and stochastic noise is injected to simulate natural variability. This formulation enables the generation of realistic subsurface patterns directly on simulation grids, avoiding the need for interpolation or geometric assumptions. Unlike traditional kriging-based workflows, which guarantee unbiasedness but often struggle to capture long-range connectivity, abrupt facies transitions, or multi-scale heterogeneity, the RD framework models heterogeneity as an emergent property of a dynamic PDE system. In contrast to classical RD-based pattern generation, which assumes fixed reaction terms, our approach derives the governing equation directly from a geostatistical objective. We begin with a Gaussian random field and define a loss function quantifying the mismatch between its empirical variogram and a target model. This loss acts as a potential energy, and the spatial field evolves by minimizing it. The variogram gradient appears as a source (reaction) term in the PDE, driving the evolution toward geostatistical consistency.

Energy Functional for Stochastic Field Generation

We model a stochastic field using a stochastic partial differential equation (SPDE):

$$\partial u(x)/\partial t = -\delta E(u)/\delta u + \sigma \eta(x, t) \quad (1)$$

Where $u(x)$ denotes a stochastic field defined on the spatial domain. In this work, $u(x)$ represents the log-permeability field, which evolves over time according to a reaction–diffusion equation. $\eta(x, t)$ represents a zero-mean, unit-variance spatiotemporal white noise field, σ controls the intensity of fluctuation (variance). This general SPDE framework has been widely studied in applied mathematics and computational modeling (Lord et al., 2014) and has also been linked explicitly to Gaussian field representations in geostatistics through the stochastic PDE approach (Lindgren et al., 2011).

The resulting SPDE is not arbitrary. It reflects an optimization dynamics with physical interpretation: The field fluctuates, but those fluctuations follow the minimization of a generalized energy functional. This equation corresponds to a Langevin equation, which describes the evolution of a system under both deterministic forces (energy gradient) and random fluctuations (noise). This noise enables exploration of multiple local minima in the energy landscape leading to diverse and structured patterns.

The field fluctuates, but those fluctuations follow the minimization of a generalized energy functional. We model the stochastic field as evolving toward low-energy configurations that still preserve variability. Physically, this reflects the idea that natural systems whether geological, chemical, or biological tend to self-organize by dissipating energy while maintaining fluctuation describes a gradient flow toward states of minimal and maximal coherence.

We define an energy functional $E(u)$ consisting of a data misfit term $J_{0(u)}$ and a regularization term:

$$E(u) = J_0(u) + \int_{\Omega} \phi(\|\nabla u\|) dx \quad (2)$$

Taking the variational derivative yields:

$$\delta E(u) / \delta u \approx -\delta J_0(u) / \delta u + \nabla \cdot \left(\int_{\Omega} \varphi(\|\nabla u\|) \frac{\nabla u}{\|\nabla u\|} dx \right) \quad (3)$$

where the grid-aware diffusion coefficient is defined as:

$$g(\|\nabla u\|) = \varphi(\|\nabla u\|) / \|\nabla u\| \quad (4)$$

Substituting Eq. (2) and (3) into the general time-evolution from (Eq. 1) leads to the reaction–diffusion equation:

$$\partial u(x) / \partial t = -\delta J_0(u) / \delta u + \nabla \cdot (g(\|\nabla u\|) \nabla u) + \sigma \eta(x, t) \quad (5)$$

The first term drives the field toward satisfying global geostatistical constraints, such as variogram or patch-based losses. The second term introduces nonlinear diffusion governed by $g(\|\nabla u\|)$ which, following the Perona–Malik formulation (Perona and Malik, 1990), can either preserve or enhance structures depending on its sign and magnitude. The final term $\eta(x, t)$ injects stochasticity, allowing diverse realizations while maintaining adherence to prior statistics.

Variogram and Histogram-Constrained Pattern Formation

To generate geologically realistic stochastic fields, we constrain the SPDE using geostatistical objectives. These constraints are encoded into the data misfit term of the previously defined energy functional. Specifically, we define the total data misfit $J_0(u)$ as a weighted sum of variogram, histogram and hard data losses:

$$J_0(u) = \lambda_v L_{\text{var}}(u) + \lambda_{\text{hist}} L_{\text{hist}}(u) + \lambda_h L_{\text{hard}}(u) \quad (6)$$

Where L_{var} penalizes deviations from a target variogram, L_{hist} penalizes deviations from true histogram, and

$$L_{\text{hard}}(u) = \sum_{x \in \Omega_h} (u(x) - u^{\text{obs}}(x))^2 \quad (7)$$

Where Ω_h is the set of hard-conditioning locations and $u^{\text{obs}}(x)$ are the observed values at those locations. The parameters λ_v , λ_{hist} and λ_h control the balance between variogram matching, histogram fidelity, and hard-data honor. Variogram-based constraints follow standard geostatistical formulations (Deutsch, 2002, Kelkar, 2002), while histogram constraints help ensure marginal distributions consistent with geological facies or petrophysical measurements (Caers, 2003).

Substituting Eqs. (6) and (7) into Eq. (5) yields the full SPDE:

$$\frac{\partial u(x)}{\partial t} = \nabla \cdot (g(\|\nabla u\|) \nabla u) + \sigma \eta(x, t) - \lambda_v \frac{\partial L_{\text{var}}(u)}{\partial u} - \lambda_{\text{hist}} \frac{\partial L_{\text{hist}}(u)}{\partial u} - \lambda_h \delta_h(x) \cdot (u(x) - u_{\text{obs}}) \quad (8)$$

Here, $\delta_h(x)$ is an indicator function equal to 1 at conditioning points and 0 elsewhere. This ensures that the simulated field honors all specified hard-data constraints exactly.

Equation (8) describes a stochastic field evolution influenced by three complementary forces:

1. Geostatistical guidance — variogram and histogram misfit terms drive the field toward target spatial correlation and marginal statistics.
2. Nonlinear diffusion — provides spatial coherence while adaptively preserving important features, controlled by $g(\|\nabla u\|)$
3. Stochastic noise — injects variability to explore multiple plausible realizations while satisfying constraints.

This combination produces fields that are geostatistically accurate, spatially coherent, and compliant with hard well data. Such integration of deterministic regularization with stochastic perturbations has parallels in both natural pattern formation (Miura & Maini, 2004) and reservoir modeling workflows (Holden et

al., 1998). This framework unifies pattern realism, hard data integration, and geological interpretability in a single SPDE-based simulation engine.

Modified Perona–Malik Diffusion for Structure Enforcement

The diffusion term in Eq. (4) plays a generative role, structuring the field spatially by encouraging smooth transitions during early evolution. In the classical Perona–Malik formulation (Perona & Malik, 1990), the diffusion coefficient decreases monotonically with gradient magnitude, suppressing smoothing across sharp transitions to preserve edges. The corresponding flux magnitude, $|J|=g(|\nabla u|) \cdot |\nabla u|$ peaks at a characteristic gradient scale λ , beyond which diffusion is strongly reduced. However, in this work we invert the traditional Perona–Malik logic. Rather than suppressing diffusion at high gradients, we amplify it in those regions to accelerate convergence toward the target variogram structure. The adopted formulation is:

$$g(r) = 1/[1 + (\lambda_0/r)^2] \quad (9)$$

where $r=|\nabla u|$ and λ_0 is a scale parameter. This function increases with r , applying stronger smoothing in regions with larger local variation. Conversely, when $r < \lambda_0$, the value of $g(r)$ is small, and diffusion becomes weak. In these nearly homogeneous regions, the field is left mostly unchanged, allowing subtle variations and low-gradient zones to persist. This selective behavior prevents over-smoothing in geologically uniform areas while directing smoothing power to zones that require structural adjustment.

Such regions of large gradients are preferentially blended with their neighbors, reducing abrupt changes and progressively aligning the spatial structure with the desired correlation range and sill. Fig 1 illustrates this behavior: the diffusion coefficient $g(r)$ increases with r , while the flux curve reaches a peak at the threshold λ_0 , highlighting the gradient scale where smoothing is strongest.

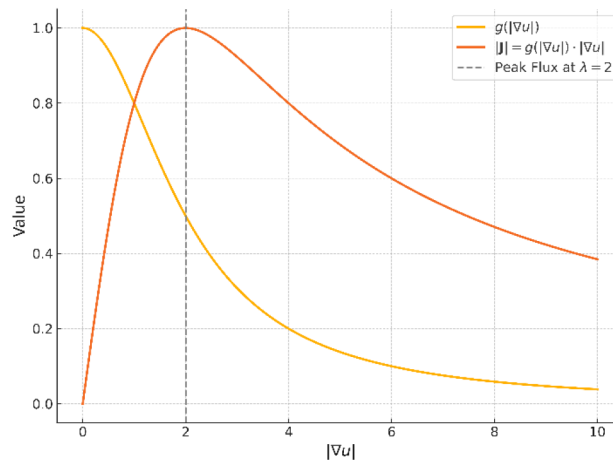


Figure 1—Perona–Malik Diffusion: selective smoothing and contrast preservation

This structure-enforcing diffusion transforms the reaction–diffusion process into a variogram-constrained smoother. During early evolution, large-gradient areas are rapidly smoothed to match the spatial continuity implied by the variogram model. As evolution proceeds, the process converges to a stable configuration where all three constraints—variogram, histogram, and hard data—are satisfied, and the generated fields exhibit geological plausibility.

This adaptive approach leverages ideas from nonlinear PDE-based image processing (Perona & Malik, 1990), pattern-forming reaction–diffusion systems (Miura & Maini, 2004), and geostatistical conditioning (Deutsch, 2002; Caers, 2003), unifying them into a single SPDE-based simulation framework for stochastic geological field generation.

Numerical experiments

We demonstrate the proposed reaction–diffusion framework through four case studies that illustrate its flexibility and geological realism in generating spatially heterogeneous permeability fields. Each case addresses a key aspect in stochastic field modeling:

Conditioning the Realization to Hard Well Data

Fig. 2 presents the starting point: a stochastic permeability field generated as a pure Gaussian-distributed initial condition, where each grid value is drawn independently with a specified mean $\log(k)$ of 5.6 and variance of 1.0. In this setup, the field has zero spatial covariance—values are uncorrelated and defined only by their mean and standard deviation—so no geological features are present at the outset. The Gaussian random field thus serves as the baseline from which spatial structure will emerge during the reaction–diffusion (RD) simulation. The left panel shows the initial condition used to initialize the reaction–diffusion (RD) process, which subsequently evolves the field over time. Black circles in the left panels indicate the locations of five wells with hard conditioning data. These well values are honored naturally during simulation by embedding them as source terms directly in the RD formulation, rather than applying separate constraints or post-processing corrections. The right panel in Fig. 2 shows a final realization that satisfies statistical constraints but does not preserve any geological streaks or barriers. Fig. 3 displays two additional realizations generated using different noise realizations, all subject to the same variogram and histogram constraints. These outcomes highlight the inherent variability in the spatial patterns generated by the stochastic PDE, despite consistent global statistics. This demonstrates that while the global structure is constrained, the local variability and feature expression remain sensitive to stochastic influences—mimicking the diversity seen in natural depositional systems.

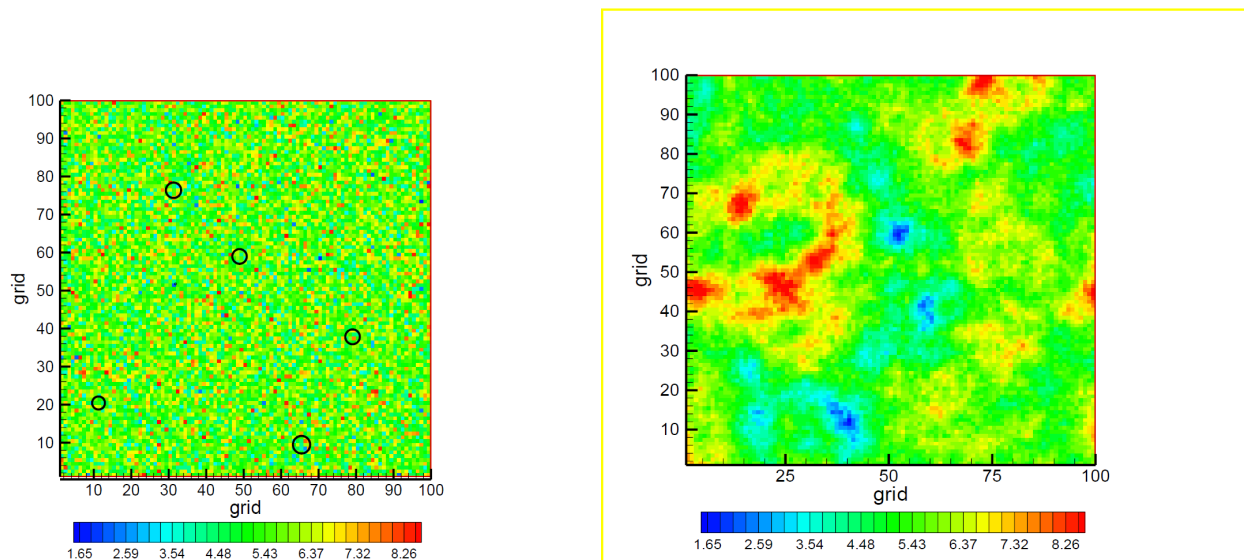


Figure 2—Log-permeability fields without streak or barrier preservation: Initial stochastic field (left) and realization (right)

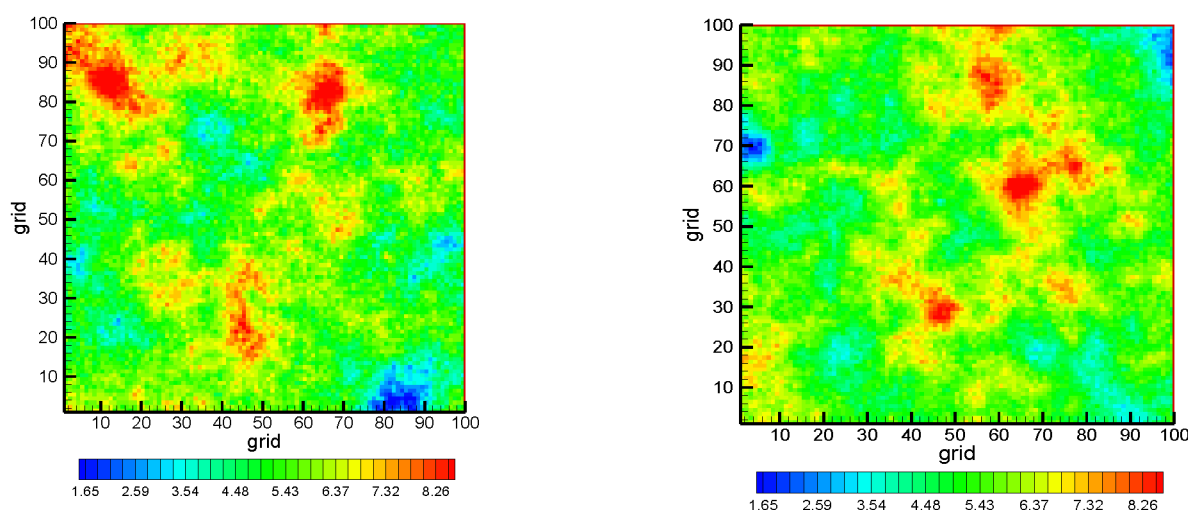


Figure 3—Two Independent realizations without streak or barrier preservation

Streak and Barrier Embedding

To preserve geological features such as streaks or facies boundaries, we apply spatial masks and modify the diffusion coefficients accordingly. Specifically, we set the diffusion coefficient $g(x)=0$ inside the streaks or barriers, effectively preventing any diffusion within these regions—mathematically equivalent to imposing a no-flow condition. However, diffusion is still allowed along the boundaries of the streaks, where adaptive values of $\lambda(x)$ in the Perona–Malik formulation control the degree of smoothing. This allows the surrounding spatial field to adjust while maintaining sharp contrasts across the streak interface. Such selective and spatially-aware smoothing is a unique capability of the reaction–diffusion framework, difficult to replicate using traditional methods such as Sequential Gaussian Simulation (SGS) or kriging, which lack intrinsic support for masked or anisotropic diffusion control.

In contrast, Fig. 4 and Fig. 5 illustrate two distinct strategies for handling geological streaks or barriers. In Figure 4, the left panel shows a Gaussian-distributed stochastic field with two cross-shaped high-permeability streaks embedded directly into the initial condition. This modified field has then evolved using the RD equation, allowing the streaks to actively influence the simulation. The right panel displays the resulting realization, where permeability is observed to cluster around the streaks—an effect caused by the interaction between the masked regions and the surrounding diffusion dynamics. In Fig. 5 the left panel shows a Gaussian-distributed stochastic field with two cross-shaped barriers, represented as low-permeability regions embedded directly into the initial condition. During the RD evolution, these barriers restrict diffusion locally but still allow limited flux along their edges. The right panel shows the final realization, where permeability clustering emerges around the barriers due to this constrained yet continuous diffusion process. The right panel displays the final realization, where low permeability clusters around the barriers—an effect caused by the restricted diffusion and leakage along the barrier boundaries. This highlights the RD model's ability to flexibly incorporate both streaks and barriers, with their influence on the surrounding field controlled through spatially adaptive diffusion properties. These two cases highlight a key advantage of the RD framework: the ability to couple pattern formation with geological constraints in a physically consistent way. Crucially, the streak regions are excluded from the computation of the variogram model, so they do not bias the statistical target. At the same time, the diffusion coefficients $g(r)$ (as defined in Equation 8) is carefully adjusted at the boundaries of the streaks, enabling smooth but controlled interaction between the abnormal zones and the surrounding field. This setup ensures that the streaks are preserved while still allowing the stochastic field to evolve coherently around them, demonstrating the flexibility and control of the RD framework in handling sharp geological features.

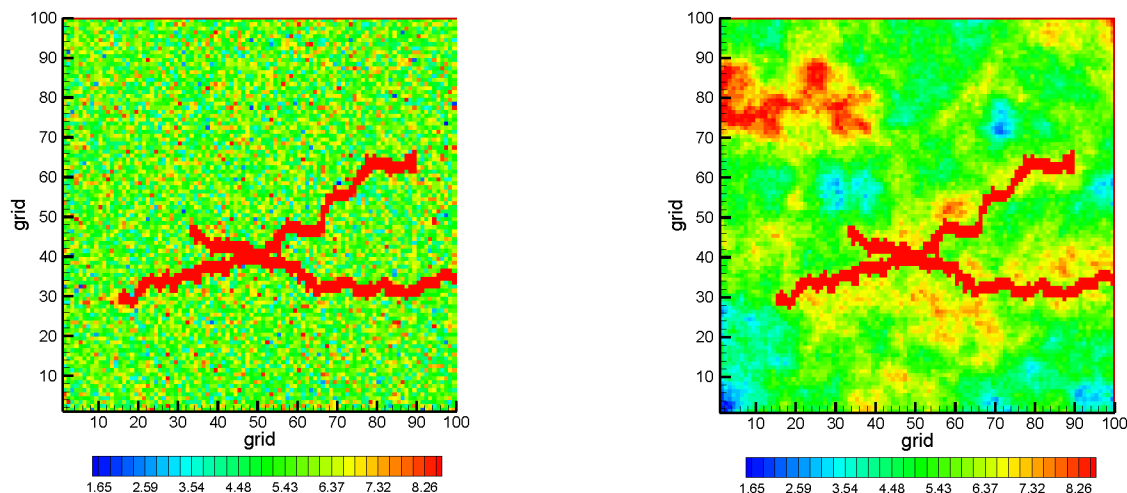


Figure 4—Log-permeability fields with streaks: Initial stochastic field (left) and realization (right)

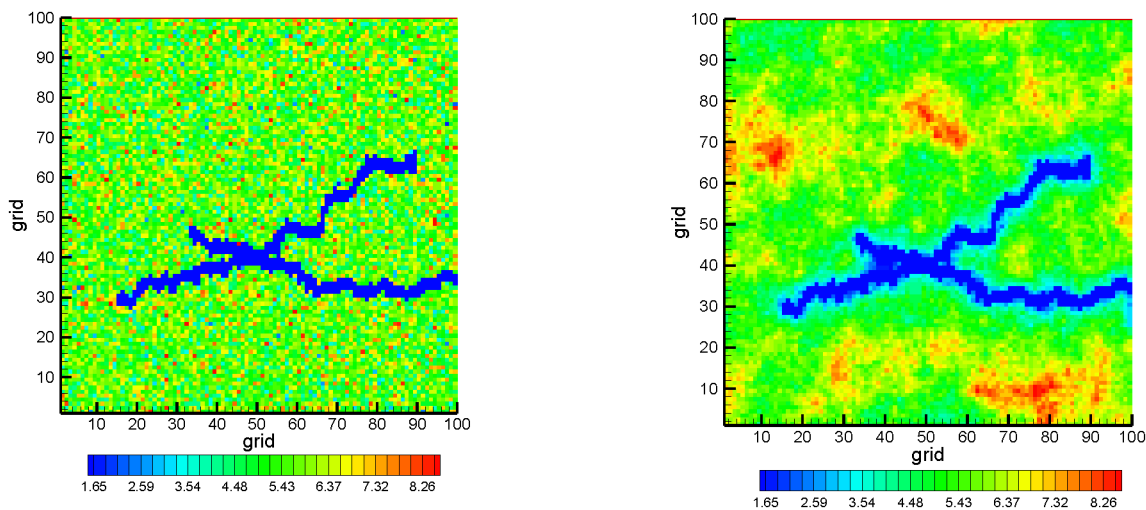


Figure 5—Log-permeability fields with barrier preservation: Initial stochastic field (left) and realization (right)

Post-Simulation Insertion

This section examines the implications of inserting geological features—high-permeability streaks or low-permeability barriers—after the RD-based stochastic simulation is complete. This approach mimics conventional object-based or SGS-style workflows, where structural features are added post hoc rather than being integrated into the generative process.

As shown in Fig. 6, the post-simulation insertion does not allow the surrounding permeability field to adapt. Unlike the embedding approach discussed earlier—where streaks are included in the initial condition and shape the diffusion evolution—features added afterward remain disconnected from their surroundings. For example, high-permeability streaks fail to induce clustering of nearby high values, and abrupt transitions appear at their boundaries, often violating variogram continuity.

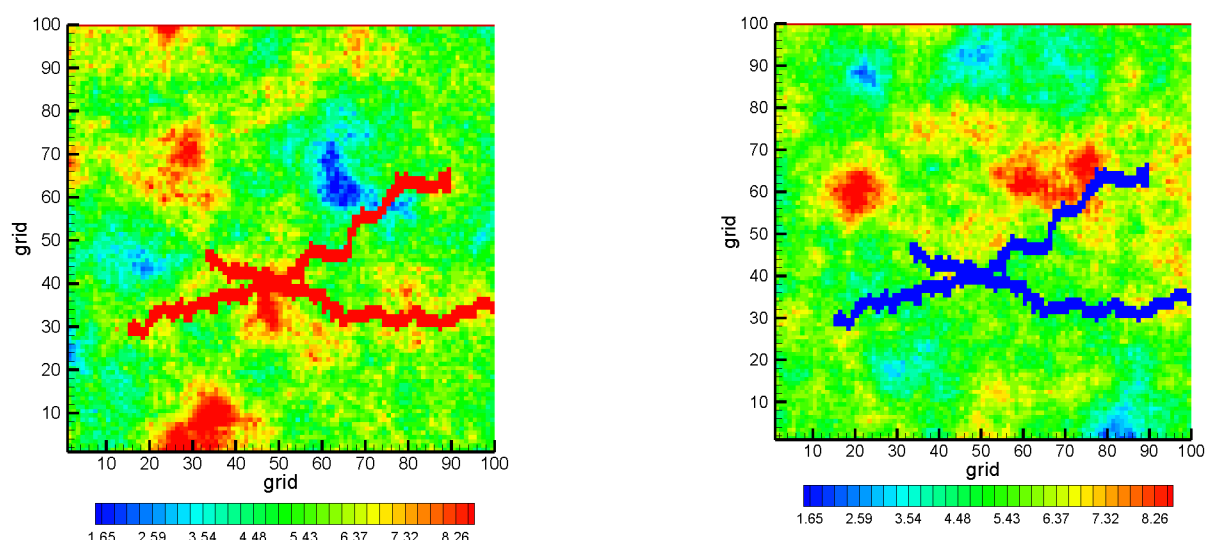


Figure 6—Log-permeability fields with inserted streaks (left) and barriers (right) after stochastic field generation

This mismatch arises because the streaks were not part of the evolution process and thus lack spatial coherence with adjacent regions. Similarly, when low-permeability barriers are inserted post-simulation, no adaptive smoothing or clustering behavior occurs around them. The barriers appear visually isolated, with no transition zone or integrated influence. These outcomes emphasize a key limitation of post-simulation insertion: the surrounding permeability field cannot reorganize around the added geological constraints. In contrast, the Reaction–Diffusion (RD) framework allows direct conditioning to streaks and barriers during simulation. This results in fields that are both geologically plausible and statistically consistent—preserving sharp contrasts where needed, while still honoring global variogram and histogram targets.

Emergence of Stochastic Field Patterns

Fig. 7 illustrates the dynamic evolution of the stochastic permeability field across six selected stages during the reaction–diffusion (RD) simulation, sampled from a total of 32 iterations leading to convergence. The simulation begins from a pure Gaussian-distributed initial condition with no imposed geological features. As the system evolves, spatial structures gradually emerge and self-organize into geologically plausible patterns. This pattern formation is driven not by explicit templates, but by the interplay between variogram constraints (which control spatial correlation across scales), histogram constraints (which preserve global distribution of values), and stochastic fluctuations injected during the simulation. As the iterations progress, fine-scale heterogeneity begins to coalesce into connected high- and low-permeability zones, mimicking realistic geological structures.

It is important to emphasize that at this stage no streaks, barriers, or hard data are present—the patterns arise solely through the internal dynamics of the RD system. The field evolves organically, balancing spatial smoothness and randomness while honoring the imposed statistical constraints. This illustrates a key strength of the RD framework: the ability to generate structured geological fields by minimizing energy functional that encodes variogram, histogram, and hard-data constraints, without relying on interpolation-based methods. The result is a field that is not only statistically consistent but also physically interpretable. The reaction–diffusion dynamics mimic a natural evolution process, where random fluctuations introduce variability and the nonlinear diffusion coefficient $g(\|\nabla u\|)$ regulates spatial organization. By adjusting this coefficient, realizations can be biased toward smoother or rougher textures, yet they remain consistent with the imposed statistical constraints. This dual role—statistical enforcement and physically motivated pattern evolution—highlights RD provides an additional option alongside traditional geostatistical approaches.

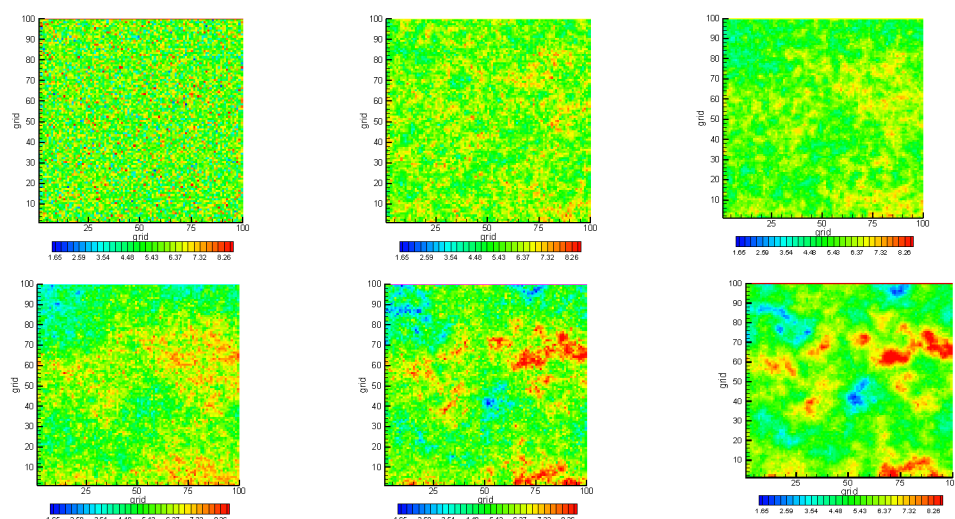


Figure 7—Selected snapshots showing the emergence of the permeability field across multiple simulation stages.

Stochastic Field on Corner-Point Grids

The proposed method is implemented directly on corner-point grids, enabling full 3D geological modeling without interpolation artifacts. The reaction–diffusion solver operates on the native simulation grid, ensuring compatibility with downstream reservoir simulation tools. The resulting realizations preserve both vertical and horizontal continuity, including features such as layer transitions and high-permeability streaks. Fig. 8 illustrates a 3D stochastic permeability field. The left panel shows the top-layer permeability with two embedded streaks, while the right panel displays the full 3D field. In 3D stochastic field generation, the actual grid dimensions (in feet) must be used rather than simple unit-based indices. Accordingly, the variogram model is computed based on physical distances between grid cells, not just their grid positions.

Importantly, the framework supports infinite realizations through variation of initial fields or noise injection. Even after a valid realization is obtained, the reaction–diffusion process can continue to produce smoother versions—useful for improving convergence in reservoir simulators. This flexibility allows users to control heterogeneity levels and generate high-quality stochastic fields aligned with physical constraints and simulation requirements.

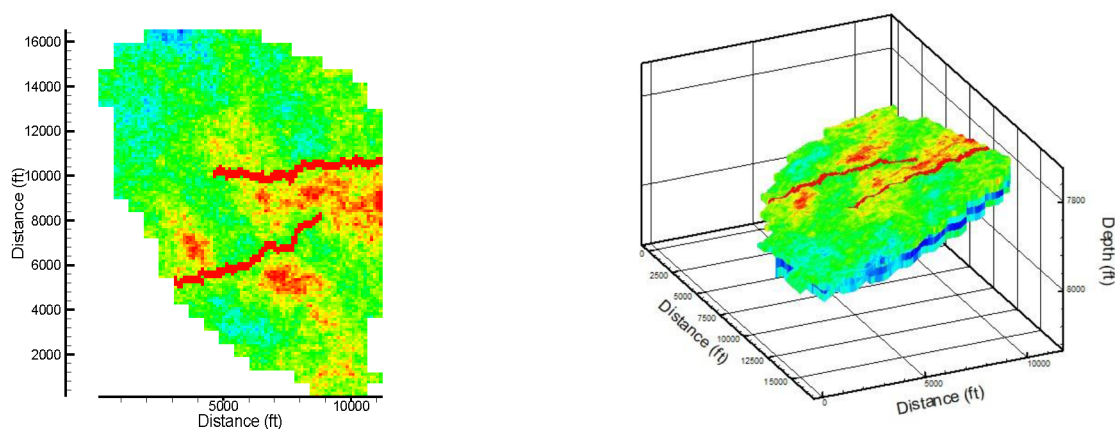


Figure 8—3D Stochastic log-permeability field on corner-point grids

Histogram and Variogram Alignment

Fig. 9 illustrates the effectiveness of the proposed RD framework in matching both variogram and histogram constraints. The left panel shows the fit between the target theoretical variogram and the empirical variogram

computed from the final realization. Initially, the variogram appears nearly flat, reflecting the unstructured nature of the Gaussian-distributed field. As the simulation evolves, the variogram adjusts dynamically and converges toward the target model, demonstrating the RD system's ability to reproduce desired spatial correlations.

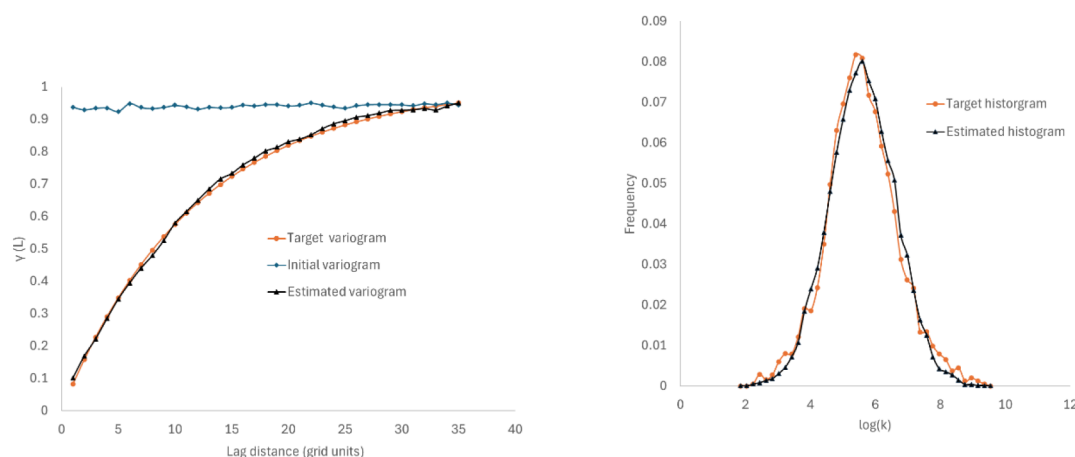


Figure 9—Histogram and variogram are fitness when the stochastic fields are generated

Unlike conventional inverse problems such as history matching or deep learning, where the initial model differs significantly from the target, the histogram in this framework is not intended to change drastically. In fact, the initial histogram is typically drawn from the target distribution, using the same mean, variance, and even random seed, so it serves more as a constraint than a mismatch to correct. The added noise during simulation introduces structural diversity without significantly distorting the global distribution. Accordingly, the optimization process is designed to preserve histogram consistency while allowing the variogram and spatial patterns to evolve. To efficiently solve this constrained variational problem, Adam is employed within the RD formulation as a robust optimizer. Originally developed for training deep learning models, Adam has recently been adapted for automatic history matching (Wu, 2025a, 2025b), and is applied here as an efficient solver for energy dissipation dynamics.

Mean Preservation in Stochastic Field Generation

Beyond spatial correlation, mean preservation is another fundamental consideration in stochastic field modeling. In ordinary kriging, unbiasedness is guaranteed by construction because the kriging weights are constrained to sum to one, ensuring that the estimated field reproduces the assumed global mean exactly. In sequential Gaussian simulation (SGS), the ensemble mean across many realizations converges to the global mean, but individual realizations may deviate—a consequence of its stochastic sampling procedure.

In the proposed RD framework, histogram constraints are explicitly enforced throughout the simulation. This ensures that the global mean and variance are preserved within each realization. Several strategies exist for mean control—such as projection (shifting the field after each step), integral constraints via Lagrange multipliers, or penalty terms—but in this work, the histogram constraint plays that role naturally, stabilizing the marginal distribution while allowing the variogram and spatial patterns to evolve.

Conclusions and Discussion

This work introduces a reaction–diffusion (RD) framework for generating geologically realistic stochastic fields directly on simulation grids. Unlike conventional geostatistical methods, such as kriging or Sequential Gaussian Simulation (SGS), which treat field generation as a statistical interpolation problem, our approach

formulates stochastic field generation as a forward simulation process governed by physically inspired dynamics.

The RD framework integrates variogram, histogram, and hard data constraints directly into a stochastic partial differential equation (PDE). By evolving from Gaussian-distributed initial conditions, the simulation captures spatial heterogeneity through a natural evolution process. The RD system progressively converges toward global statistical targets while maintaining geological realism.

Hard data constraints, including well observations, are incorporated directly as spatial source terms in the reaction term. This built-in conditioning mechanism enables the model to honor control points without requiring post-processing or artificial smoothing, in contrast to SGS-style workflows. The RD system adapts dynamically to these local conditions while preserving continuity across the domain.

A notable strength of the RD framework lies in its ability to integrate geological features—such as high-permeability streaks or barriers—into the simulation. When these structures are embedded before the simulation, via the initial condition and spatially masked diffusion coefficients, they influence the evolution of surrounding parameter fields, inducing directional clustering and preserving edge continuity. In contrast, inserting such features after simulation typically leads to unrealistic visual discontinuities and edge inconsistencies. This comparison underscores the RD model's capacity to enforce structural realism through its governing equations.

Across all experiments, the generated fields closely match the prescribed variogram and histogram statistics, even in the presence of conditioning data or complex features. The ability to selectively halt or modulate diffusion within specific zones (by setting $g = 0$ inside streaks) gives fine-grained control over geological plausibility while still meeting statistical constraints.

From a practical standpoint, the RD framework enables a paradigm shift in reservoir modeling workflows. Instead of building a parameter field and later mapping it to the simulation grid, one can define the simulation grid first—corner-point, unstructured, or otherwise—and evolve the property fields within it. This inversion of traditional practice aligns better with physics-based simulation and facilitates downstream integration with flow modeling, uncertainty quantification, and history matching.

While reaction–diffusion systems have been extensively studied in biology, chemistry, physics, and materials science—including applications in animal patterning, tissue morphogenesis, 3D printing, water filtration, and functional material design—their use in geosciences remains largely unexplored. The present work introduces RD dynamics as a complementary tool for reservoir modeling, showing that concepts originally developed to explain biological patterns or engineer advanced materials can also generate geologically realistic heterogeneity directly on simulation grids.

In conclusion, the reaction–diffusion approach provides a flexible and physically grounded alternative to classical geostatistical techniques. By unifying stochasticity, spatial regularization, and statistical objectives in a single PDE-based framework, the method offers a powerful and extensible option for generating realistic geological parameter fields directly on simulation grids. Future work may extend this framework to support stochastic anisotropic field generation for advanced subsurface modeling applications.

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Appendix A

Gradients of Variogram and Histogram Loss Functions

Variogram-Based Loss and Its Gradient

To incorporate spatial correlation constraints, we match the empirical variogram of the simulated field to a target variogram. The variogram loss function is defined as:

$$J_{Vario} = \sum_{h=1}^{N(h)} \left(\gamma_{sim}(h) - \gamma_{target}(h) \right)^2 \quad (A-1)$$

where $\gamma(h)$ is the empirical variogram at lag h , defined by:

$$\gamma(h) = \frac{1}{N(h)} \sum_{(i,j,k) \in Q_h} \left(u(i, j, k) - u(i', j', k') \right)^2 \quad (A-2)$$

where: Q_h is the set of grid pairs separated by lag h , $N(h)$ is the number of such grid pairs

The gradient of the variogram loss with respect to the field $u(i, j, k)$ is given by:

$$\frac{\partial J_{Vario}}{\partial u(i, j, k)} = \sum_{h=1}^H \frac{2}{N(h)} \left(\gamma_{sim}(h) - \gamma_{target}(h) \right) \cdot \left(u(i, j, k) - u(i', j', k') \right) \quad (A-3)$$

This expression accounts for each grid's participation in all directional lag-pairings. Symmetric terms from reverse lag pairs $(i-h, j, k)$ can be included similarly to ensure consistency.

Histogram-Based Loss and Its Gradient

To enforce the target frequency distribution of property values, we define a histogram-based loss that penalizes mismatch between the simulated and reference histograms:

$$J_{hist} = \sum_{b=1}^N \left(hist_{sim}(b) - hist_{target}(b) \right)^2 \quad (A-4)$$

where $hist_{sim}(b)$ is the normalized frequency in bin b computed from the current field, $hist_{target}(b)$ is the reference histogram or initial conditions and N is the total number of histogram bins.

The gradient of this loss with respect to each grid parameter u is computed by distributing contributions across nearby histogram bins based on relative distance in value space:

$$\frac{\partial J_{hist}}{\partial u(i, j, k)} = \sum_{b=1}^n \frac{2}{n} \left(hist_{sim}(b) - hist_{target}(b) \right) \cdot w_b(i, j, k) \quad (A-5)$$

where $w_b(i, j, k)$ represents the sensitivity of bin b to changes in $u_{i,j,k}$, decreasing smoothly with distance between the parameter value and the bin center.